MILITARY SPECIFICATION SHEET

STARTER, AIRCRAFT ENGINE, HYDRAULIC, FOR T64 ENGINE

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the starter described herein shall consist of this specification and the latest issue of MIL-S-22999.

REQUIREMENTS:

Envelope: Figures 1 and 2

Wt, max dry (lb): 20.0

Aircraft: Model CH-53

Engine: Model T64

Quick-attach-detach mounting: Meet requirements of MS14118(AS)

Engine accessory drive: The starter shall be suitable for mounting on an AND20002, Type II-S drive, except that the stud location shall be changed to position two studs on the vertical center line.

Rotation viewed from anti-output shaft end: Clockwise

Ports: Conform to: Inlet - AND1OO5O-10; Outlet - MS33649-12

Ambient temperature operating range (“F): -65 to +275

Exposure temperature range (“F): -103 to +275

Overspeed limit, min peak no-load speed (rpm): 8000

Inlet pressure at zero fluid flow, max (psig): 4400

Rated inlet pressure (psig): 3750 +50, -0

Rated outlet pressure (psig): 5O +15, -O

FSC 29

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Rated inlet fluid temperature (“F): +275 +0, -10

Rated fluid flow, max (gpm): 21.5; maximum allowable flow at no-load operation is 23.0 gpm.

Rated torque (lb-ft): 47 at 3700 psig differential pressure

Rated drive shaft speed (rpm): 4000

Automatic shutoff control: The cutout switch shall be normally closed; it shall open at cutout speed. The switch shall withstand a surge current of 2 amperes at 28 volts inductive load at -65°F.

Drive shaft cutout speed (rpm): 6700 ±150

Initial calibration:
Hydraulic fluid flow, max (gpm):
Zero to max cutout drive speed: 21.5
No-load drive speed: 23.0

Performance:
Minimum Starter Drive Torque (lb-ft)
and Corresponding Starter Drive Speeds (rpm)

Pressure Differential (PSIG)	Speed	Torque	Speed	Torque	Speed	Torque	Speed	Torque
3700	0	37.5	1000	50	4000	47	6000	30

No-load operation: The starter shall be operated for two minutes. The starter free-running speed obtained during this test, or the maximum speed attained before automatic cutback, shall not be less than 8000 rpm.

Proof pressure test: With the load device adjusted to limit starter output drive speed to rated speed rpm and at rated inlet temperature, the starter shall be operated for five minutes at 5000-psig inlet pressure and 1250 psig outlet pressure. There shall be no leakage from the starter sufficient to form a drop, except from the output shaft seal from which a maximum leakage of 5 cc per hour is allowable.

Endurance test: The starter shall be subjected to a 1200 cycle endurance test as outlined herein. The following schedule for fluid inlet temperature and fluid inlet pressure shall be maintained:

Number of Cycles	Inlet Fluid Temperature (“F)	Inlet Fluid Pressure (psig)
400	275	3750
600	220	3900
200	160	3750

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All tests shall be conducted at rated outlet pressure.
Each cycle shall consist of three consecutive phases as follows:

Test Stand Rotor Polar Moment of Inertia	Phase A - Acceleration Drive		Phase B - 10 sec at Following Conditions		Phase C - Acceleration Drive	
	Speed RPM	Time	Drive Speed RPM	Load Torque	Speed RPM	Time
Lb-Ft²	(Min)	Sec	(Min)	Lb-Ft	(Min)	Sec
47.2	2450	11	2450	47 +3-O	7150	13

Note: The above values do not allow for flywheel windage losses, test stand drag, or inertia of the starter. In order to determine compliance with this specification, the test equipment must be calibrated and specific test results corrected accordingly. If the acceleration specified for Phase C is not developed because of test stand drag torque, the test stand inertia load may be reduced or external power may be applied to assist acceleration to cutout speed. Time for end of Phase C to starter cutout shall not exceed 15 seconds.

Sustained motoring: Operational time, six minutes

Overrunning test: The starter output shaft shall be driven for 1000 hours in a clockwise direction (viewed from the anti-output shaft end) in accordance with the following tabulation:

Starter Drive Overrunning Speed (rpm)

Phase A 750 Hours	Phase B 200 Hours	Phase C 50 Hours
12,200	12,400	9,050

After a maximum of each 25 hours of continuous overrunning, output shaft rotation shall be stopped for at least 15 minutes.

Low temperature operation: During Phase B of this test at a drive torque of 45 lb-ft, a minimum drive speed of 2450 rpm shall be developed.

Dynamic torque check: The maximum peak transient torque shall be 200 1b-ft.

Shear section strength, max static torque (lb-ft): 200

Shaft seal leakage (cc per hour): 1 at 12,400 rpm overrunning speed

Over pressurization pressure (psig): 4500

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Part number: The part number shall consist of the basic number of this specification sheet.

Example: M22999/2

Preparing Activity
Navy - AS
(Project No.2995-N153)

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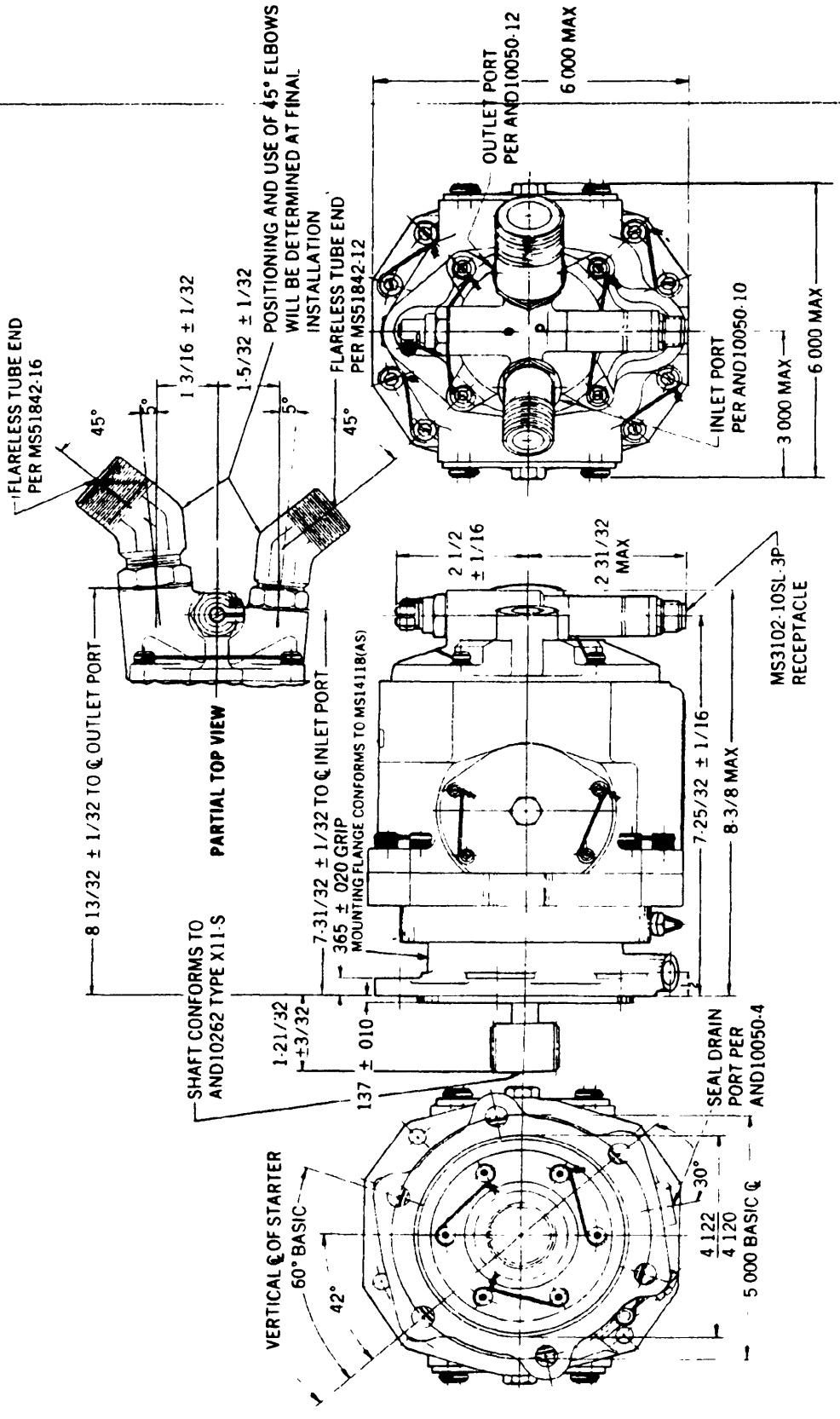


FIGURE 1. Starter envelope

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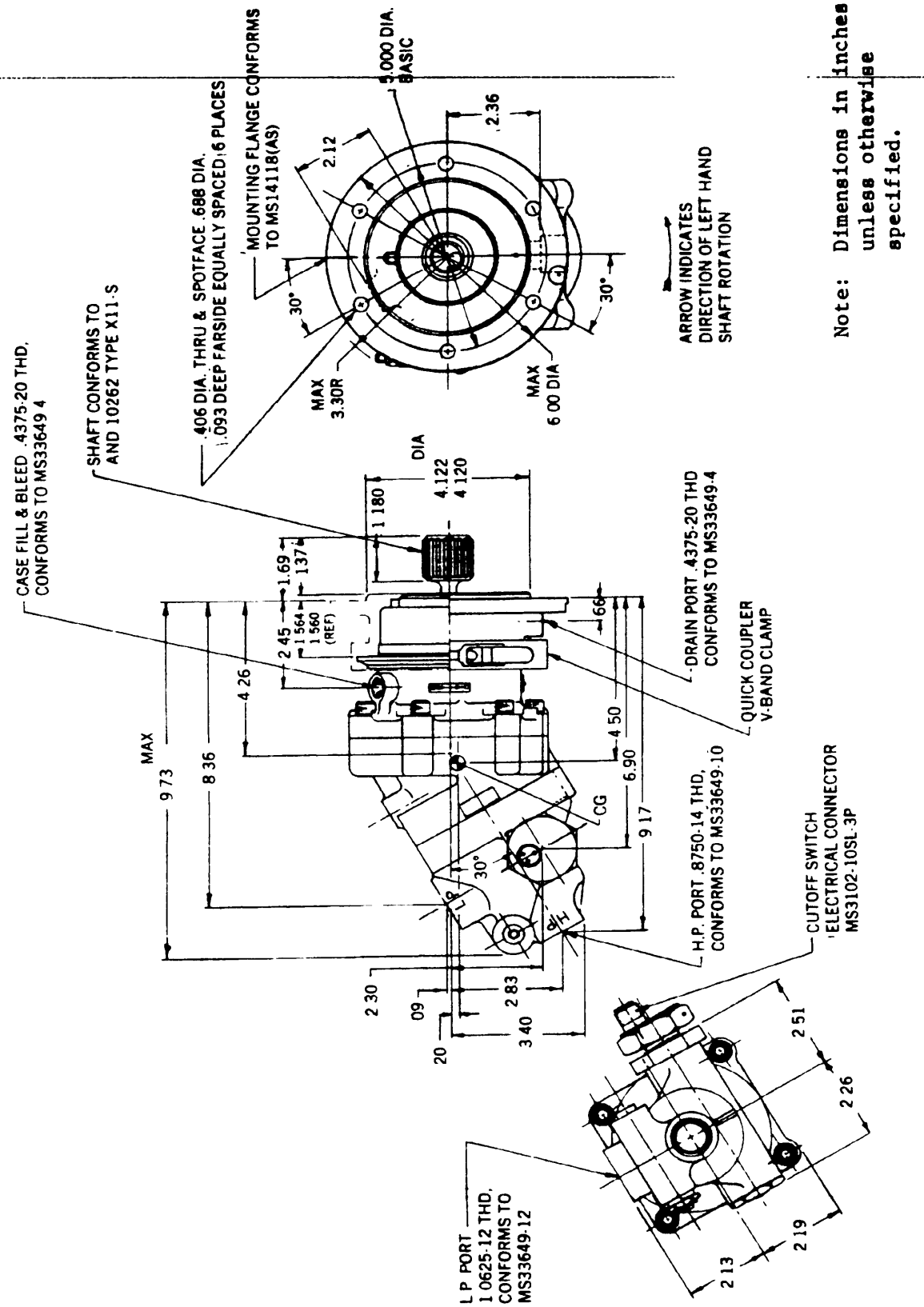


FIGURE 2. Starter envelope (alternate configuration)

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